

Difference-in-Differences with Multiple Time Periods*

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Abstract

In this article, we consider identification, estimation, and inference procedures for treatment effect parameters using Difference-in-Differences (DiD) with (i) multiple time periods, (ii) variation in treatment timing, and (iii) when the “parallel trends assumption” holds potentially only after conditioning on observed covariates. We show that a family of causal effect parameters are identified in staggered DiD setups, even if differences in observed characteristics create non-parallel outcome dynamics between groups. Our identification results allow one to use outcome regression, inverse probability weighting, or doubly-robust estimands. We also propose different aggregation schemes that can be used to highlight treatment effect heterogeneity across different dimensions as well as to summarize the overall effect of participating in the treatment. We establish the asymptotic properties of the proposed estimators and prove the validity of a computationally convenient bootstrap procedure to conduct asymptotically valid simultaneous (instead of pointwise) inference. Finally, we illustrate the relevance of our proposed tools by analyzing the effect of the minimum wage on teen employment from 2001–2007. Open-source software is available for implementing the proposed methods.

JEL: C14, C21, C23, J23, J38.

Keywords: Difference-in-Differences, Dynamic Treatment Effects, Doubly Robust, Event Study, Variation in Treatment Timing, Treatment Effect Heterogeneity, Semi-Parametric.

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adoption in the literature. We interpret this assumption as if units do not “forget” about the treatment experience.⁴

Define G as the time period when a unit first becomes treated. Under Assumption 1, for all units that eventually participate in the treatment, G defines which “group” they belong to. If a unit does not participate in any time period, we arbitrarily set $G = \infty$. We define G_g to be a binary variable that is equal to one if a unit is first treated in period g (i.e., $G_{i,g} = \mathbf{1}\{G_i = g\}$) and define C to be a binary variable that is equal to one for units that do not participate in the treatment in any time period (i.e., $C_i = \mathbf{1}\{G_i = \infty\} = 1 - D_{i,\mathcal{T}}$). Let $\bar{g} = \max_{i=1,\dots,n} G_i$ be the maximum G in the dataset. Next, denote the generalized propensity score as $p_{g,s}(X) = P(G_g = 1 | X, G_g + (1 - D_s)(1 - G_g) = 1)$. Note that $p_{g,s}(X)$ indicates the probability of being first treated at time g , conditional on pre-treatment covariates X and on either being a member of group g (in this case, $G_g = 1$) or a member of the “not-yet-treated” group by time s (in this case, $(1 - D_s)(1 - G_g) = 1$). Many of our results use a specialized version of this generalized propensity score, and, henceforth, we define $p_g(X) = p_{g,\mathcal{T}}(X) = P(G_g = 1 | X, G_g + C = 1)$ which is the probability of being first treated in period g conditional on covariates and either being a member of group g or not participating in the treatment in any time period. Let $\mathcal{G} = \text{supp}(G) \setminus \{\bar{g}\} \subseteq \{2, 3, \dots, \mathcal{T}\}$ denote the support of G excluding $\bar{g}$.⁵ Likewise, let $\mathcal{X} = \text{supp}(X) \subseteq \mathbb{R}^k$ denote the support of the pre-treatment covariates. Finally, for a generic $\delta \geq 0$, let $\mathcal{G}_\delta = \mathcal{G} \cap \{2 + \delta, 3 + \delta, \dots, \mathcal{T}\}$.

Next, we set up the potential outcomes framework. Here, we combine the dynamic potential outcomes framework of Robins (1986, 1987) with the multi-stage treatment adoption setup discussed by Heckman et al. (2016); see also Sianesi (2004). Let $Y_{i,t}(0)$ denote unit i ’s untreated potential outcome at time t if they remain untreated through time period $\mathcal{T}$; i.e., if they were not to participate in the treatment across all available time periods. For $g = 2, \dots, \mathcal{T}$, let $Y_{i,t}(g)$ denote the potential outcome that unit i would experience at time t if they were to first become treated in time period g . Note that our potential outcomes notation accounts for potential dynamic treatment selection, though it also accommodates (pre-specified) treatment regimes (Murphy et al., 2001; Murphy, 2003). The observed and potential outcomes for each unit i are related through

$$Y_{i,t} = Y_{i,t}(0) + \sum_{g=2}^{\mathcal{T}} (Y_{i,t}(g) - Y_{i,t}(0)) \cdot G_{i,g} \quad (2.1)$$

In other words, we only observe one potential outcome path for each unit. For those that do not participate in the treatment in any time period, observed outcomes are untreated potential outcomes in all periods. For units that do participate in the treatment, observed outcomes are the unit-specific potential outcomes corresponding to the particular time period when that unit adopts the treatment.

We also impose the following random sampling assumption.

Assumption 2 (Random Sampling). $\{Y_{i,1}, Y_{i,2}, \dots, Y_{i,\mathcal{T}}, X_i, D_{i,1}, D_{i,2}, \dots, D_{i,\mathcal{T}}\}_{i=1}^n$ is independent and identically distributed (*iid*).

Assumption 2 implies that we have access to panel data; our results extend essentially immediately

⁴See Han (2020), de Chaisemartin and D’Haultfoeulle (2020) and Bojinov et al. (2020) for alternative setups where treatment can “turn off”.

⁵When there is a “never treated” set of units with $G = \infty$, $\mathcal{G}$ only excludes this group. When such “never-treated” group is not available, we exclude the latest-treated group as there will be no valid untreated comparison group for them.